

Tree-Level Calibration Is Not Plot-Level Calibration: Uncertainty Propagation Pitfalls in Drone-Based Agroforestry Carbon Estimation

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Abstract

Drone-based measurement, reporting and verification (MRV) is lowering the cost of bringing smallholder agroforestry into carbon markets, but the step that actually issues credits — turning a diameter (DBH) model’s predictions into a conservative plot-level claim — is rarely validated. On open ground-truth data (6,292 trees, semi-arid western India) we show that this step can fail even when the DBH model looks reliable. Per tree, predictive intervals are near nominal (93% coverage at a nominal 90% under a tree-level split; 86–90% on held-out farms). Per plot, they are strongly over-confident: nominal 80% intervals contain the field-measured total for only 46–62% of pseudo-plots, consistent with errors that are shared within a plot instead of averaging out. Claiming a posterior lower quantile (p_{25}) therefore over-claims on 46–58% of plots, and a flat 80% haircut over-claims on 19–35% while forgoing 13–17% of credits in aggregate. Holding out whole plots makes this worse: nominal 80% intervals then cover only 12–36% of plots, and p_{25} over-claims on 49–61%. Separately, collapsing each tree’s prediction to a point before the convex allometry lowers plot totals by 14–17%, as the classical lognormal correction predicts. These results hold across DBH models from a pooled Gaussian process to hierarchical Bayesian models. Species-aware partial pooling reduces per-tree DBH error by 13–17%, but only when species is known, and it does not reduce aggregate CO₂ bias. We close with a checklist for practitioners.

Keywords: carbon MRV; agroforestry; uncertainty quantification; calibration; hierarchical Bayesian models; Gaussian processes

1. Introduction

Tree planting on smallholder farms combines climate mitigation with income diversification for farmers in climate-vulnerable regions. Carbon finance could pay for it, but only if the carbon stored can be measured, reported and verified (MRV) at a cost proportional to small, fragmented plots. Field inventories do not meet that bar. Drone imagery combined with machine learning does: a detector finds trees in an orthomosaic, crown size and height are read from the imagery and a digital elevation model, a regression model predicts each tree’s diameter at breast height (DBH), and an allometric equation converts DBH into biomass and CO₂ (Gupta et al., 2025).

Most of the attention in such pipelines goes to the learned components: detection and DBH regression. This paper is about what happens *after* the DBH model — how its uncertainty is propagated to a plot total and turned into a credit claim. Those steps are a few lines of code each, but they decide how many credits are issued, and errors in them are invisible in the per-tree accuracy metrics that model development usually tracks.

We study three questions on an open agroforestry ground-truth dataset ([Farmers for Forests, 2025](#)), using up to seven DBH models with different covariates and inductive biases, so that each conclusion can be checked for model dependence:

1. **Back-transform.** How large is the bias from pushing a point estimate of $\log(\text{DBH})$ through a convex allometry, and is it model-dependent?
2. **Plot-level calibration.** Do intervals that are near-nominal per tree remain calibrated for plot totals, which is the unit at which credits are issued? How do common issuance policies — a flat fractional haircut, or a posterior lower quantile — behave at the plot level?
3. **Covariates.** How much of the accuracy gain from species-aware hierarchical modelling survives when only drone-observable covariates are available?

Contributions. (i) A same-split comparison of seven DBH models with matched metrics, including a covariate ablation separating what is deployable from imagery alone (§3). (ii) Empirical confirmation across all seven models that the point-estimate back-transform under-reports plot CO_2 by 14–17%, matched by the closed-form lognormal correction (§4). The correction is classical in forest allometry ([Baskerville, 1972](#); [Sprugel, 1983](#)); our point is that it re-enters ML pipelines whenever the upstream model’s predictive distribution is collapsed to a point. (iii) A pseudo-plot evaluation of five models, under both a tree-level split and a split that holds out whole plots, showing that near-nominal per-tree calibration does not transfer to plot totals, so that neither a flat haircut nor a raw posterior quantile is a safe issuance rule (§5). (iv) A short practitioner checklist (§6).

2. Setting, Data and Models

2.1. From DBH to credits

We use the allometric chain of the open drone-MRV methodology of [Gupta et al. \(2025\)](#): a multi-species power law for above-ground biomass (AGB, kg) taken from Verra-listed literature — Eq. 3.2.1 of [Brown \(1997\)](#), fitted on 28 trees from dry forest in India; total biomass adds roots (root-to-shoot ratio 0.27); carbon is 50% of dry mass; and CO_2 is 44/12 of carbon, following the Verra VMD0001 module ([Verra, 2023](#)):

$$\text{AGB} = e^a \text{DBH}^b, \quad a = -1.996, \quad b = 2.32, \quad \text{CO}_2 = 1.27 \times 0.5 \times \frac{44}{12} \times \text{AGB}. \quad (1)$$

All pipelines in this paper predict $\log(\text{DBH})$, as is standard for allometric quantities. A plot’s claimed CO_2 is then a functional of the predictive distribution of the plot total $T = \sum_{i \in \text{plot}} \text{CO}_{2i}$. We compare five issuance policies: the *point* total (each tree’s median DBH pushed through Eq. 1); the *mean* of T under per-sample propagation; the posterior lower quantiles $p_{25}(T)$ and $p_{10}(T)$; and a flat *80% rule*, $0.8 \mathbb{E}[T]$, the conservative policy recommended in an open drone-MRV methodology ([Gupta et al., 2025](#)).

2.2. Data

We use the public ground-truth release of 13 October 2025 ([Farmers for Forests, 2025](#)): 7,326 field-measured trees from four districts of Maharashtra, India, each with species, DBH, height, two crown widths, plantation year, field-recorded age, block and GPS location. We drop implausible records ($\text{DBH} \leq 1$ or ≥ 200 cm, height ≤ 0.5 m, crown ≤ 0.3 m, $\text{DBH} < 2$ cm with height > 3 m) and records without age; exclude bamboo (72 records), to which the hardwood allometry does not apply; and keep species with at least 30 trees. The modelling set has **6,292 trees from 18 species** (median DBH 8.0 cm, range 1.3–53.5 cm). It is geographically concentrated: 92% of trees are in one district. We use an 80/20 split stratified by species (5,035 / 1,257 trees); every model in this paper is trained and scored on this same split.

Pseudo-plots. The release has no plot identifier. To evaluate plot-level behaviour we group test trees into pseudo-plots in two ways: *block-year* (district $\times$ block $\times$ plantation year; 26 groups with ≥ 10 test trees, covering 1,208 trees, median 25 trees per group) and *geo-cell* (GPS rounded to 10^{-3} degrees, ≈ 110 m; 39 groups, 894 trees, median 20). Block-year is our primary grouping; geo-cell checks robustness.

2.3. DBH models

Pooled Gaussian process (GPR). Following the specification of the open methodology above ([Gupta et al., 2025](#)), we fit a GP ([Rasmussen and Williams, 2006](#)) on standardised crown size and height with a scaled RBF kernel plus white noise, target $\log(\text{DBH})$, using scikit-learn ([Pedregosa et al., 2011](#)). A variant on log inputs, and a pooled log-log ordinary least squares (OLS) model, serve as further pooled baselines.

Hierarchical Bayesian family. Writing $\tilde{x}$ for a log-covariate centred on its training mean, the full model (*H-full*) is

$$\begin{aligned} \log(\text{DBH}_i) &\sim \mathcal{N}(\mu_i, \sigma_{s_i}^2), \\ \mu_i &= \alpha_{s_i} + \beta_{s_i} \tilde{c}_i + \zeta_{s_i} \tilde{c}_i^2 + \gamma_{s_i} \tilde{h}_i + \eta_{s_i} m_{s_i} \tilde{a}_i + \delta_{d_i}, \end{aligned} \tag{2}$$

with s_i the species, d_i the district, c, h, a crown, height and age. Species coefficients are partially pooled ([Gelman and Hill, 2007](#)) towards global means in non-centred form; the age slope is constrained non-negative ($\eta_s = \eta_g + |\xi_s|$) so that older trees are not predicted thinner; $m_s = 0$ masks age for the two species whose within-species age coefficient of variation is below 0.10. Priors are listed in Appendix A. Three ablations switch terms off without other changes:

- *H-drone*: species pooled ($\alpha, \beta, \zeta, \gamma$ shared), district effect kept, no age — deployable from imagery plus location;
- *H-species*: species-level coefficients, no district, no age;
- *H-species+district*: species and district, no age.

We sample with NUTS ([Hoffman and Gelman, 2014](#)) in PyMC ([Abril-Pla et al., 2023](#)): 4 chains $\times$ 1,000 draws after 1,000 tuning steps, target acceptance 0.95. All runs have

Table 1: Per-tree accuracy on the same 1,257 test trees. “Drone-only” marks models whose inputs are observable from imagery plus location. Calibration slope is the slope of empirical against nominal coverage at levels 0.5–0.95 (1 is ideal; < 1 means under-confident).

Model	Inputs	Drone-only	RMSE (cm)	MAE (cm)	Cal. slope
GPR	crown, height	yes	2.87	1.75	0.86
GPR (log inputs)	crown, height	yes	2.67	1.73	0.86
Pooled OLS	crown, height	yes	2.84	1.86	0.89
H-drone	+ district	yes	2.83	1.81	0.87
H-species	+ species	no	2.34	1.57	0.90
H-species+district	+ species, district	no	2.39	1.57	0.89
H-full	+ species, district, age	no	2.24	1.48	0.88

$\hat{R} \leq 1.01$ on every hyperparameter and no divergent transitions. All predictive distributions are represented by 1,000 joint samples over the test set; for the GPR we sample from the full joint predictive covariance, so that plot totals carry correlated error in every model.

3. What Covariates Buy

Table 1 separates the gain from the hierarchical structure from the gain from the covariates it can use. With drone-observable inputs only, the hierarchical model (H-drone, 2.83 cm) is indistinguishable from the pooled GP (2.67–2.87 cm) and OLS; the district effect adds nothing. Knowing the species reduces RMSE to 2.34 cm (a 17% reduction over H-drone), and field-recorded age reduces it a further 6% to 2.24 cm. District adds nothing on top of species.

On held-out farms (§5.1) the picture narrows. RMSE is 3.31 cm for GPR, 3.36 cm for H-drone, 2.94 cm for H-species (13% below H-drone), 3.16 cm for H-species+district and 2.79 cm for H-full. The species gain survives on the DBH scale but not on the log scale on which the models are fitted (MAPE on $\log(\text{DBH})$ is 10.4–10.9% for all five models), and the district effect now hurts, consistent with district effects learned on one set of farms not transferring to new ones. The practical reading is that partial pooling over species buys a modest gain in DBH accuracy *if* species can be identified from imagery; until then, the simpler pooled model is as accurate. Per tree, all models are mildly under-confident under the tree-level split (90% intervals contain $\approx 93\%$ of test trees) and close to nominal on held-out farms (86–90%).

Per-tree accuracy is not the quantity that credits are issued on. The species-aware models without age over-predict the test-set CO_2 total by 9.2% under per-sample propagation, against 2–3% for the pooled models and 4.5% for H-full; on held-out farms the pooled GPR has the least biased total (+0.3%, against +3.5 to +16.9% for the hierarchical models). Lower per-tree error does not imply a less biased total. We return to plot totals in §5.

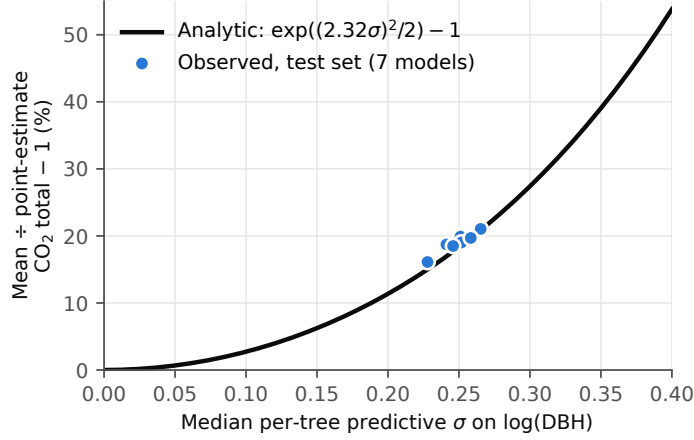


Figure 1: Gap between the per-sample mean and the point-estimate CO_2 total on the test set, for seven models, against each model’s median per-tree predictive σ . The curve is Eq. 3.

4. The Back-Transform Gap

If $\log(\text{DBH}) \sim \mathcal{N}(\mu, \sigma^2)$, then by Eq. 1

$$\frac{\mathbb{E}[\text{AGB}]}{e^a (e^\mu)^b} = \exp\left(\frac{(b\sigma)^2}{2}\right) = \exp(2.69\sigma^2) \quad (b = 2.32). \quad (3)$$

A pipeline that collapses each tree’s predictive distribution to its median (or to e^μ) before applying the allometry therefore under-reports biomass by this factor, whatever the model. The correction is well known for allometric regressions fitted in log space (Baskerville, 1972; Sprugel, 1983; Sileshi, 2014); the same bias re-appears in ML pipelines where σ is the *predictive* uncertainty of an upstream model and the allometry is applied downstream as a fixed post-processing step. Analytical approximations that linearise the allometry around the mean (the delta method) drop the same term.

Figure 1 and Table 2 show the effect on the test set. For every model the point-estimate total is 14–17% below the per-sample mean (gap 16–21% of the point total), and Eq. 3 at the model’s median σ predicts the gap to within 2 percentage points. The point-estimate totals land 8–15% *below* ground truth. A pipeline that makes this choice is already conservative by roughly the size of a typical haircut, before any haircut is applied — and a practitioner comparing models by point-estimate totals would see the best per-tree model (H-full) as biased low by 10% when its per-sample total is within 5%. The fix requires no retraining: propagate samples, or apply Eq. 3 per tree.

5. Tree-Level Calibration Does Not Transfer to Plots

Credits are issued per plot, so the quantity that must be calibrated is the predictive distribution of the plot total T (Gneiting et al., 2007). We evaluate the five models of §3 that have

Table 2: Test-set CO₂ total relative to field ground truth under the point-estimate and the per-sample (mean) back-transform, and the gap between them against the analytic prediction at the model’s median per-tree σ .

Model	Point	Mean	Gap (observed)	Median σ	Gap (Eq. 3)
GPR	−14.9%	+2.0%	19.9%	0.251	18.5%
GPR (log inputs)	−13.5%	+2.9%	19.0%	0.252	18.6%
Pooled OLS	−14.8%	+3.2%	21.0%	0.265	20.9%
H-drone	−12.1%	+5.2%	19.7%	0.258	19.7%
H-species	−8.0%	+9.2%	18.7%	0.241	16.9%
H-species+district	−7.8%	+9.2%	18.5%	0.246	17.7%
H-full	−10.0%	+4.5%	16.1%	0.228	15.0%

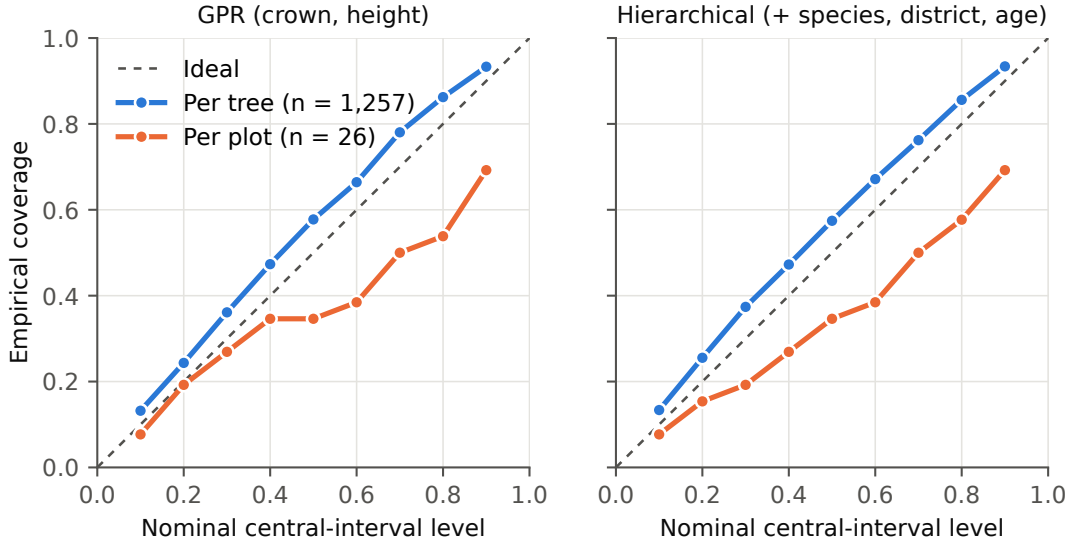


Figure 2: Empirical coverage of central predictive intervals for individual trees (blue) and for block-year pseudo-plot totals (orange). Both models are slightly under-confident per tree and strongly over-confident per plot.

full joint predictive samples (GPR and the four hierarchical models) on the pseudo-plots of §2.2, first under the tree-level split used so far, then with whole plots held out (§5.1). Figure 2 compares coverage per tree and per plot for the pooled GPR and H-full. Per tree, both models are slightly under-confident. Per plot, both are strongly over-confident: nominal 80% intervals contain the true total for 54% (GPR) and 58% (H-full) of block-year plots. Across the five models and both groupings, 80% plot intervals cover 46–67% of plots and 50% intervals cover 19–44% (Appendix B).

The reason is visible in the per-plot errors. For H-full, the mean-policy error on block-year plots ranges from −35% to +55%, and its magnitude is uncorrelated with plot size

Table 3: Issuance policies on 26 block-year pseudo-plots (tree-level split). Over-claim: share of plots where the claim exceeds the field-measured total, with 95% Wilson intervals. Left / over: total under- and over-shoot of the 80% rule across plots, as a share of true CO₂.

Model	Over-claim rate, % [95% CI]				80% rule
	mean	p_{25}	p_{10}	80% rule	left / over
GPR	65 [46–81]	46 [29–65]	31 [17–50]	23 [11–42]	16.7 / 1.4
H-drone	62 [43–78]	54 [35–71]	42 [26–61]	31 [17–50]	15.7 / 4.0
H-species	77 [58–89]	58 [39–74]	31 [17–50]	19 [9–38]	13.3 / 1.2
H-species+district	77 [58–89]	54 [35–71]	42 [26–61]	35 [19–54]	14.3 / 3.6
H-full	65 [46–81]	46 [29–65]	31 [17–50]	19 [9–38]	14.5 / 1.5
Nominal target	—	25	10	—	

(Spearman $\rho = -0.07$). Errors that were independent between trees would shrink with plot size; these do not. They are shared within a plot — plausibly through management (orchard pruning), site conditions or measurement crews — and no model in this study has a term for them. Summing per-tree predictive distributions that are conditionally independent given the global parameters then yields a plot-total distribution that is far too narrow.

Consequences for issuance. Table 3 scores the policies plot by plot. A posterior lower quantile is an attractive replacement for a flat haircut, since it adapts the deduction to each plot’s uncertainty and has a clear interpretation — but only if the plot-level distribution is calibrated. Here it is not: p_{25} over-claims on 46–58% of plots instead of 25%, and p_{10} on 31–42% instead of 10%. With 26 plots the intervals are wide, but for every model the 95% interval excludes the nominal rate for both quantiles. The flat 80% rule is more conservative in aggregate — it forgoes 13–17% of true CO₂ — yet still over-claims on 19–35% of plots, because plot errors larger than 20% are common. The geo-cell grouping gives the same picture (p_{25} over-claims on 41–54%, the 80% rule on 20–41%; Appendix B). The hierarchical model with all covariates has the smallest plot-level error (mean absolute error 14.7% under p_{25} , against 19.2% for GPR), so covariates help, but they do not fix the calibration.

Neither policy is safe, for the same reason: both treat the model’s tree-level uncertainty as if it described the plot. The remedy is to calibrate at the level of issuance — for example a plot-level random effect fitted on surveyed plots, or a conformal or empirical quantile adjustment estimated on held-out plots (Angelopoulos and Bates, 2023) — and to check over-claim rates against the nominal level on plots the model has not seen.

5.1. Held-out plots

The tree-level split places trees from every plot in both training and test sets, which may flatter plot-level performance. We therefore repeat the evaluation with whole block-year groups held out: 5-fold grouped cross-validation over the 35 groups in the full modelling set, refitting all five models in each fold (GPR plus four hierarchical models; 20 MCMC

Table 4: Plot-held-out evaluation (5-fold grouped cross-validation, 33 held-out plots). Tree cov.: share of held-out trees inside their 90% interval. Plot cov.: share of plots whose true total lies in the central 50% / 80% interval. Over-claim rates with 95% Wilson intervals. Left / over: 80% rule, as in Table 3.

Model	Tree cov.	Plot cov.	Over-claim rate, % [95% CI]			80% rule
	90%	50 / 80%	p_{25}	p_{10}	80% rule	left / over
GPR	90%	6 / 12%	49 [33–65]	49 [33–65]	30 [17–47]	24.9 / 5.5
H-drone	90%	12 / 30%	58 [41–73]	46 [30–62]	33 [20–50]	24.1 / 7.5
H-species	87%	18 / 36%	61 [44–75]	46 [30–62]	30 [17–47]	17.4 / 10.5
H-species+district	86%	18 / 33%	55 [38–70]	46 [30–62]	33 [20–50]	23.1 / 9.9
H-full	87%	12 / 33%	49 [33–65]	36 [22–53]	27 [15–44]	19.7 / 4.9
Nominal target	90%	50 / 80%	25	10	—	

fits). Test trees whose species is absent from a fold’s training set are dropped (40 trees); a district absent from training receives a zero district effect. This yields 33 held-out plots with at least 10 trees (median 119, range 25–926) and 6,252 held-out trees. All fits have $\hat{R} \leq 1.01$; three species-level fits in two folds have divergent transitions on at most 1.3% of draws.

Table 4 shows that the tree-level split did flatter the models. Per tree, coverage on unseen farms is still close to nominal (86–90% at 90%). Per plot, nominal 80% intervals now contain the true total for only 12–36% of plots, against 46–62% under the tree-level split, and 50% intervals for 6–18%. p_{25} over-claims on 49–61% of plots and p_{10} on 36–49%; for every model the 95% interval again excludes the nominal rate. The 80% rule over-claims on 27–33% of plots while leaving 17–25% of true CO₂ unclaimed. Extrapolation also exposes a fragility of the *mean* policy: on one held-out plot — a 1990 planting in the smallest district, far older than most training plots — the H-species posterior mean over-predicts CO₂ ten-fold because a few extreme predictive draws dominate the average, while its p_{25} is off by 33%. Quantile-based claims are more robust to such tails, but, as above, not calibrated.

6. Discussion

Checklist for practitioners.

1. *Propagate samples, not points.* Push predictive samples of $\log(\text{DBH})$ through the full allometric chain and sum per sample; or apply Eq. 3 per tree. Compare totals between models only after doing so.
2. *Validate at the level of issuance.* Report coverage and over-claim rates of plot totals on held-out plots, not per-tree coverage.
3. *Model shared plot error.* Include a plot- or farm-level effect, or calibrate the plot-total quantile empirically, before using it for credits.

4. *Split by plot.* Tree-level random splits place trees from the same farm in training and test, which flattens plot-level error: here, holding out plots cut 80% plot-interval coverage from 46–62% to 12–36%.
5. *Separate deployable from field covariates.* Report accuracy with the inputs available at inference time.

Limitations. Our plots are pseudo-plots built from administrative and spatial groupings, not surveyed plots, and there are few of them (26, 39 and 33), so rates carry wide intervals (Tables 3 and 4). The held-out evaluation removes farm overlap between training and test, but held-out groups remain administrative units, and the smallest districts are represented by only a few plots. The data come from one semi-arid region (92% from one district) and contain only two test trees above 50 cm DBH. Ground truth is itself the allometry applied to field DBH, so allometric error (Chave et al., 2014) and tree-detection error are outside our scope. The allometry is also used outside its stated domain: it was fitted for DBH 5–40 cm and is recommended for dry zones with more than 900 mm/year of rainfall (Brown, 1997), whereas 92% of our trees come from a district with a normal annual rainfall of 561.6 mm (ICAR-CRIDA, 2011). This affects absolute CO₂ levels but not our conclusions. Re-running the full evaluation with the companion equation that Brown (1997) recommends below 900 mm ($AGB = 10^{-0.535} BA$, i.e. $b = 2$; Appendix C) shrinks the back-transform gap to 12–14%, as Eq. 3 predicts, while p_{25} still over-claims on 46–54% of block-year plots and nominal 80% plot intervals still cover only 46–65%. Saplings (<5 cm, 15% of test trees) carry about 1% of CO₂ but have large relative measurement noise.

Conclusion. The uncertainty-handling steps after the DBH model — back-transform, aggregation and issuance — matter as much as the model. On open agroforestry data, the choice of back-transform moves plot totals by 14–17% regardless of model, and intervals that are near-nominal per tree under-cover plot totals by a wide margin, which makes both flat haircuts and naive posterior quantiles unreliable issuance rules. Both checks are cheap and should be part of any drone-based MRV pipeline.

Code and data. Code, saved posteriors and per-plot results: <https://anonymous.4open.science/r/Carbon-MRV-Audit-5F10/>. The ground-truth data are the public release of Farmers for Forests (2025), used without modification.

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Appendix A. Priors

Global means: $\alpha_g \sim \mathcal{N}(2.5, 0.5^2)$, $\beta_g \sim \mathcal{N}(0.5, 0.3^2)$, $\zeta_g \sim \mathcal{N}(0, 0.2^2)$, $\gamma_g \sim \mathcal{N}(0.3, 0.3^2)$, $\eta_g \sim \mathcal{N}^+(0.2, 0.3^2)$ (truncated at 0). Species scales: $\tau_\alpha \sim \mathcal{HN}(0.5)$, $\tau_\beta \sim \mathcal{HN}(0.3)$, $\tau_\zeta \sim \mathcal{HN}(0.15)$, $\tau_\gamma \sim \mathcal{HN}(0.3)$, $\tau_\eta \sim \mathcal{HN}(0.2)$. Species coefficients (non-centred): $\alpha_s = \alpha_g + \tau_\alpha z_{\alpha,s}$ and likewise for β, ζ, γ , with $z \sim \mathcal{N}(0, 1)$; age slope $\eta_s = \eta_g + \xi_s$, $\xi_s \sim \mathcal{HN}(\tau_\eta)$. District effects $\delta_d \sim \mathcal{N}(0, \tau_\delta^2)$, $\tau_\delta \sim \mathcal{HN}(0.3)$. Observation noise $\sigma_s \sim \mathcal{HN}(0.5)$ per species (a single σ when species is pooled). Ablations drop the corresponding terms and priors.

Appendix B. Robustness: Geo-Cell Pseudo-Plots and Coverage

Table 5: Over-claim rate by policy on 39 geo-cell pseudo-plots (≈ 110 m cells), and central-interval coverage of plot totals for both groupings (block-year / geo-cell).

Model	Over-claim rate (geo-cell)				Plot coverage	
	mean	p_{25}	p_{10}	80% rule	50%	80%
GPR	77%	46%	36%	28%	35 / 41%	54 / 59%
H-drone	72%	54%	44%	41%	19 / 26%	46 / 46%
H-species	72%	44%	26%	28%	23 / 44%	62 / 67%
H-species+district	67%	46%	26%	28%	27 / 38%	50 / 67%
H-full	67%	41%	28%	21%	35 / 38%	58 / 62%

Appendix C. Robustness: Low-Rainfall Allometry

Table 6: The evaluation of §4–5 repeated with Eq. 3.2.2 of [Brown \(1997\)](#) ($\text{AGB} = 10^{-0.535} \text{BA}$, BA in cm^2 ; $b = 2$), recommended for dry zones below 900 mm/year, against the main-text allometry (Eq. 3.2.1, $b = 2.32$). Block-year pseudo-plots. Gap: per-sample mean over point-estimate total, minus one.

Model	Gap		Over-claim rate, $b = 2$			80% coverage
	$b = 2.32$	$b = 2$	p_{25}	p_{10}	80% rule	$b = 2.32 / b = 2$
GPR	19.3%	13.9%	46%	31%	19%	54 / 58%
H-drone	19.8%	14.3%	54%	42%	27%	46 / 46%
H-species	18.6%	13.3%	46%	27%	15%	62 / 65%
H-species+district	18.5%	13.2%	54%	38%	27%	50 / 54%
H-full	16.0%	11.5%	46%	23%	15%	58 / 65%
Nominal target	—	—	25%	10%	—	80%